Cisco Packet Tracer Activity Summary

Two-Router Network Configuration

Overview

This activity involved setting up a two-router network in Cisco Packet Tracer with the goal of successfully sending messages between PCs on opposite sides of the network. The setup involved configuring router interfaces, establishing a serial link between routers, adding static routes, and assigning IP addresses to all PCs.

Network Topology

The network consists of two routers (R1 and R2) connected via a serial link, each with two switches and multiple PCs:

Side	Router	Switch	PCs	Network
R1 Side	R1 (1941)	SW1	PC1, PC2, PC3	`192.168.10.0/24`
R1 Side	R1 (1941)	SW2	PC4, PC5, PC6	`192.168.11.0/24`
R2 Side	R2 (1941)	SW3	PC7, PC8, PC9	`10.1.1.0/24`
R2 Side	R2 (1941)	SW4	PC10, PC11, PC12	`10.1.2.0/24`

Cable / Connector Types

Cable Type	Used For
Console (RS-232)	Laptop to Switch — for initial configuration
Straight-through (CST)	PC to Switch — standard connection
Crossover (CCO)	Switch to Switch — connecting same device types
Serial (Red line)	Router to Router — WAN link between R1 and R2

Step 1: Configure R1 Interfaces

Access R1 via CLI tab and enter the following commands:

```
enable
configure terminal
```

```
! G0/0 - connects to SW1 (192.168.10.x network)
interface gigabitethernet 0/0
ip address 192.168.10.1 255.255.255.0
no shutdown
exit

! G0/1 - connects to SW2 (192.168.11.x network)
interface gigabitethernet 0/1
ip address 192.168.11.1 255.255.255.0
no shutdown
exit

! S0/0/0 - serial link to R2
interface serial 0/0/0
ip address 209.165.200.225 255.255.255.252
no shutdown
exit
```

Step 2: Configure R2 Interfaces

Access R2 via CLI tab and enter the following commands:

```
enable
configure terminal

! G0/0 - connects to SW3 (10.1.1.x network)
interface gigabitethernet 0/0
ip address 10.1.1.1 255.255.255.0
no shutdown
exit

! G0/1 - connects to SW4 (10.1.2.x network)
interface gigabitethernet 0/1
ip address 10.1.2.1 255.255.255.0
no shutdown
exit

! S0/0/0 - serial link to R1
interface serial 0/0/0
ip address 209.165.200.226 255.255.255.252
no shutdown
```

```
exit
```

Step 3: Add Static Routes

Static routes tell each router how to reach networks on the other side. Without these, routers cannot forward traffic between R1 and R2 networks.

On R1 — tell R1 how to reach R2's networks:

```
ip route 10.1.1.0 255.255.255.0 209.165.200.226
ip route 10.1.2.0 255.255.255.0 209.165.200.226
exit
```

On R2 — tell R2 how to reach R1's networks:

```
ip route 192.168.10.0 255.255.255.0 209.165.200.225
ip route 192.168.11.0 255.255.255.0 209.165.200.225
exit
```

Syntax explanation: ip route [network to reach] [subnet mask] [send traffic to this IP]

Step 4: Assign IP Addresses to All PCs

For each PC: Click PC → Desktop → IP Configuration → Select Static → Fill in the values below.

R1 Side PCs:

PC	IPv4 Address	Subnet Mask	Default Gateway
PC1	192.168.10.10	255.255.255.0	192.168.10.1
PC2	192.168.10.11	255.255.255.0	192.168.10.1
PC3	192.168.10.12	255.255.255.0	192.168.10.1
PC4	192.168.11.10	255.255.255.0	192.168.11.1
PC5	192.168.11.11	255.255.255.0	192.168.11.1
PC6	192.168.11.12	255.255.255.0	192.168.11.1

R2 Side PCs:

PC	IPv4 Address	Subnet Mask	Default Gateway
PC7	10.1.1.10	255.255.255.0	10.1.1.1

PC8	10.1.1.11	255.255.255.0	10.1.1.1
PC9	10.1.1.12	255.255.255.0	10.1.1.1
PC10	10.1.2.10	255.255.255.0	10.1.2.1
PC11	10.1.2.11	255.255.255.0	10.1.2.1
PC12	10.1.2.12	255.255.255.0	10.1.2.1

Step 5: Test with Ping

From any PC on R1 side → Desktop → Command Prompt:

```
ping 10.1.1.10     ! reach PC7 on R2 side
ping 10.1.2.10     ! reach PC10 on R2 side
```

Expected result — success looks like:

```
Reply from 10.1.1.10: bytes=32 time=33ms TTL=126
```

Note: The first packet timing out (Request timed out) is normal. This happens because the router needs a moment to resolve ARP (MAC addresses) on the very first packet. After that, all replies succeed.

Router Interface Summary

Router	Interface	IP Address	Subnet Mask	Connects To
R1	G0/0	192.168.10.1	255.255.255.0	SW1 (PC1-PC3)
R1	G0/1	192.168.11.1	255.255.255.0	SW2 (PC4-PC6)
R1	S0/0/0	209.165.200.225	255.255.255.252	R2 Serial Link
R2	G0/0	10.1.1.1	255.255.255.0	SW3 (PC7-PC9)
R2	G0/1	10.1.2.1	255.255.255.0	SW4 (PC10-PC12)
R2	S0/0/0	209.165.200.226	255.255.255.252	R1 Serial Link

Quick Command Reference

Command	What It Does
enable	Enter privileged EXEC mode

configure terminal	Enter global configuration mode
interface gigabitethernet 0/0	Access a specific interface
ip address [IP] [mask]	Assign IP address to interface
no shutdown	Turn the interface on
ip route [net] [mask] [next-hop]	Add a static route
show ip route	View routing table
show ip interface brief	Check all interface statuses
exit	Go back one configuration level

Result

✅ Successfully configured a two-router network in Cisco Packet Tracer.

✅ PCs on the R1 side (192.168.x.x) can communicate with PCs on the R2 side (10.1.x.x).

✅ All ping tests passed with 3/4 replies (first packet ARP delay is normal).

✅ Messages successfully sent across both networks using the simulation tool.